Hussain Singapuri

AI ENGINEER (GENAI, RAG AND AGENTIC AI)

+91 9561277305 | Hussainsingapuri24@gmail.com | linkedin.com/in/hussain-singapuri-18437225a | github.com/huss2003 | hussain-singapuri.vercel.app | Pune, India

PROFESSIONAL SUMMARY

AI Engineer with 3 years of professional experience at a US-based medical-device company, where I built **AI agents** for internal use alongside full-stack web applications and a CRM. Builds **GenAI** applications in **Python**: **RAG** with hybrid BM25 + **vector** retrieval (pgvector), **multi-agent orchestration** with tool use, retries and token budgets, **LLM API** integrations (Claude, Gemini), evals and **prompt** injection defence, plus an **LLM inference** engine with 3.1x throughput. Works with **Git**, **GitHub Actions CI/CD**, **Docker**, **REST APIs** and automated testing. Claude Certified Developer; Google Cloud Generative AI Leader.

CORE COMPETENCIES

Agentic AI & Multi-Agent Orchestration | GenAI & RAG Architectures | Vector Databases (pgvector) | LLM APIs & Prompt Engineering | Python & FastAPI | LLM Evaluation & Guardrails | LLM Inference & Latency Optimisation | Git, CI/CD & Containerization (Docker)

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an **AI tutor**: phased spec, server sizing, engineering team plan and CAPEX/OPEX model
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)
- Built and run the website neuronetwrks.com with a **Python** static-site generator deployed on Vercel, including programmatic SEO
- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built **AI agents** for internal use
- Built several apps, including a **CRM**, and maintained full-stack web applications for the company
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

multi-agent-research-crew

Open Source

Multi-agent orchestration: planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

site-to-agent

Open Source

RAG: turns any website into a grounded chat agent with a crawler, hybrid BM25 + vector retrieval with RRF, cited answers, SSRF-hardened fetching and an eval harness.

Next.js, TypeScript, pgvector

nanoserve Open Source

LLM inference engine from scratch: paged KV cache, prefix caching, continuous batching with preemption, speculative decoding, OpenAI-compatible streaming API; 3.1x throughput over sequential decoding in benchmarks.

Python, PyTorch, FastAPI

PageForge Portfolio project

LLM application with guardrails: AI drafts only from approved facts, a deterministic validator flags unsupported numbers, a reviewer signs off claim by claim, engagement flows to analytics and the CRM.

Next.js 16, React 19, Sanity, Supabase, Gemini, PostHog, HubSpot

ai-lead-receptionist Open Source

AI assistant for home-service businesses: missed-call text-back, SMS and web chat that qualifies leads and books jobs, with an owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

Safey Inventory

Traceability and production system for a small manufacturing unit: forward and backward batch tracing, FIFO allocation, low-stock alerts, batch recall, planning and reporting.

TanStack Start, React 19, Tailwind v4, shadcn/ui, TanStack Query

More projects

contractor-ai-ops-dashboard, lead-intel-engine, Orbital Surplus, Personal portfolio with full-stack backend, Nightjar & Co., TripSync, Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

Pune

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

AI / LLM: AI agents, multi-agent orchestration, tool use / function calling, RAG (hybrid BM25 + vector, RRF), evals, prompt-injection defence, LLM inference serving, Claude API, Gemini API, Claude Code and AI coding agents

Languages: Python, TypeScript, JavaScript, SQL, Dart, HTML, CSS

Data: pgvector, PostgreSQL, Neon, Supabase (RLS, Edge Functions), PGlite, SQLite, Sanity CMS

Back end: FastAPI, asyncio, REST APIs, Node.js, webhooks (HMAC signing), auth (scrypt, sessions, throttling), Stripe Checkout and refunds, Resend

DevOps: Git, GitHub Actions CI/CD, Docker (containerization), Vercel

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Front end: Next.js, React 19, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO